

i want you to teach me how to use django forms
by making a demo project and an application with a
templates folder.

so that i could learn,

add data to a table / model using forms.
and to bring data to form inorder to edit it.

database is in postgres name : demo db

employee model having columns : employee id, name,
salary,email,mobile number

1. Create Project

Bash

```
django-admin startproject demoProject
cd demoProject
python manage.py startapp employeeApp
```

2. Install PostgreSQL Driver

Bash

```
pip install psycopg2
```

3. Configure PostgreSQL

Open:

Python

Run

```
demoProject/settings.py
```

Replace DATABASES with:

Python

Run

```
DATABASES = {
    'default': {
        'ENGINE': 'django.db.backends.postgresql',
        'NAME': 'demo_db',
        'USER': 'postgres',
        'PASSWORD': 'your_password',
        'HOST': 'localhost',
        'PORT': '5432',
    }
}
```

4. Register App

In settings.py

```
Python

INSTALLED_APPS = [
    ...
    'employeeApp',
]
```

Run

5. Create Model

Open:

```
Python

employeeApp/models.py
```

Run

```
Python

from django.db import models

class Employee(models.Model):

    employee_id = models.CharField(max_length=20, unique=True)
    name = models.CharField(max_length=100)
    salary = models.DecimalField(max_digits=10, decimal_places=2)
    email = models.EmailField()
    mobile_number = models.CharField(max_length=15)

    def __str__(self):
        return self.name
```

Run

6. Make Migrations

Bash

```
python manage.py makemigrations
python manage.py migrate
```

7. Create Forms

Open:

Python

Run

```
employeeApp/forms.py
```

Python

Run

```
from django import forms
from .models import Employee

class EmployeeForm(forms.ModelForm):

    class Meta:
        model = Employee

        fields = [
            'employee_id',
            'name',
            'salary',
            'email',
            'mobile_number'
        ]
```

8. Create Views

Open:

Python

Run

```
employeeApp/views.py
```

Python

Run

```
from django.shortcuts import render, redirect, get_object_or_404
```

```
from .models import Employee
from .forms import EmployeeForm


# INSERT DATA
def add_employee(request):

    if request.method == "POST":

        form = EmployeeForm(request.POST)

        if form.is_valid():
            form.save()
            return redirect('employee_list')

    else:
        form = EmployeeForm()

    return render(request, 'add_employee.html', {'form': form})


# SHOW DATA
def employee_list(request):

    employees = Employee.objects.all()

    return render(request, 'employee_list.html', {
        'employees': employees
    })


# EDIT DATA
def edit_employee(request, id):

    employee = get_object_or_404(Employee, id=id)

    if request.method == "POST":

        form = EmployeeForm(request.POST, instance=employee)

        if form.is_valid():
            form.save()
            return redirect('employee_list')

    else:
        form = EmployeeForm(instance=employee)

    return render(request, 'edit_employee.html', {
        'form': form
    })
```

9. Create URLs

employeeApp/urls.py

Python

Run

```
from django.urls import path
from . import views

urlpatterns = [

    path('', views.employee_list, name='employee_list'),

    path('add/', views.add_employee, name='add_employee'),

    path('edit/<int:id>/', views.edit_employee, name='edit_employee'),
]
```

10. Main URLs

Open:

Python

Run

demoProject/urls.py

Python

Run

```
from django.contrib import admin
from django.urls import path, include

urlpatterns = [
    path('admin/', admin.site.urls),

    path('', include('employeeApp.urls')),
]
```

11. Create Templates Folder

Inside:

Python

Run

employeeApp/

Create:

templates/

12. add_employee.html

HTML



```
<!DOCTYPE html>
<html>
<head>
  <title>Add Employee</title>
</head>
<body>

  <h1>Add Employee</h1>

  <form method="POST">

    {% csrf_token %}

    {{ form.as_p }}

    <button type="submit">Save</button>

  </form>
```

13. employee_list.html

HTML



```
<!DOCTYPE html>
<html>
<head>
  <title>Employee List</title>
</head>
<body>

  <h1>Employees</h1>

  <a href="/add/">Add Employee</a>

  <table border="1" cellpadding="10">

    <tr>
      <th>ID</th>
      <th>Employee ID</th>


```

```
<th>Name</th>
```

14. edit_employee.html

HTML



```
<!DOCTYPE html>
<html>
<head>
  <title>Edit Employee</title>
</head>
<body>

<h1>Edit Employee</h1>

<form method="POST">

  {% csrf_token %}

  {{ form.as_p }}

  <button type="submit">Update</button>

</form>
```

15. Run Server

Bash

```
python manage.py runserver
```

16. What You Learned

INSERT DATA

Python

Run

```
form = EmployeeForm(request.POST)

if form.is_valid():
    form.save()
```

SHOW EXISTING DATA IN FORM

Python

Run

```
form = EmployeeForm(instance=employee)
```

This fills form fields automatically.

UPDATE EXISTING DATA

Python

Run

```
form = EmployeeForm(request.POST, instance=employee)
```

This updates existing row instead of creating new row.

Flow Understanding

Add Employee

```
HTML Form
↓
request.POST
↓
EmployeeForm
↓
form.save()
↓
PostgreSQL Table
```

Edit Employee

```
Existing Object
↓
instance=employee
↓
Form Prefilled
↓
User Changes Data
↓
form.save()
↓
Updated Row
```



what is it? raise
ImproperlyConfigured(msg.format(name=self.urlconf_name)) from e
django.core.exceptions.ImproperlyConfigured: The included URLconf '`<module 'employeeApp.urls' from 'C:\\Users\\Lenovo\\Desktop\\demopro\\demoProject\\employeeApp\\urls.py'>`' does not appear to have any patterns in it. If you see the 'urlpatterns' variable with valid patterns in the file then the issue is probably caused by a circular import.

This means Django opened:

Python

Run

```
employeeApp/urls.py
```

but could not find a valid:

Python

Run

```
urlpatterns = [  
]
```

OR your file has an import problem.

Most Common Reasons

1. urlpatterns typo

Check carefully:

Python

Run

```
urlpatterns = [  
]
```

NOT:

Python

Run

```
url_patterns
urlpatternsss
urlpatterns
```

Correct File

Your `employeeApp/urls.py` should be:

Python

Run

```
from django.urls import path
from . import views

urlpatterns = [

    path('', views.employee_list, name='employee_list'),

    path('add/', views.add_employee, name='add_employee'),

    path('edit/<int:id>/', views.edit_employee, name='edit_employee'),
]
```

2. Syntax Error in views.py

If `views.py` has an error,
then `urls.py` fails while importing views.

Then Django thinks no `urlpatterns` exist.

Check terminal carefully for earlier errors.

3. Circular Import

Example bad situation:

Python

Run

```
views.py imports urls.py
```

and

Python

Run

```
urls.py imports views.py
```

Do NOT import `urls` inside `views.py` .

4. Indentation Error / Missing Bracket

Very common.

Example:

Python

```
urlpatterns = [  
    path('', views.home)
```

Run

Missing:

Python

```
]
```

Run

Quick Debug

Temporarily replace your `employeeApp/urls.py` with:

Python

```
from django.urls import path  
from django.http import HttpResponse  
  
def test(request):  
    return HttpResponse("Working")  
  
urlpatterns = [  
    path('', test),  
]
```

Run

If server works now,
problem is inside `views.py` .



Is this conversation helpful so far?

